

1. Which of the following represent data and information respectively?
 - a. Unit price of an item and daily sales report
 - b. Body Mass Index and Height
 - c. Heat and Temperature
 - d. Weight and growth of plants
2. Which of the following is/are true about CPU?
 - A. CPU does decision making and data processing.
 - B. ALU is responsible for executing arithmetic and logical operations.
 - C. Machine language is not CPU dependent.

a. A and C b. A and B c. None of the above d. All of the above
3. Which option(s) represents the main components of the Von Neumann Architecture?
 - a. ALU, Control Unit, instruction, input device, data
 - b. Data, ALU, output device, Control unit, main memory
 - c. Main memory, CPU, ALU, control unit, output device
 - d. Data, address, CPU, main memory, ALU
4. According to the Stored Program Concept, where are the programs to be processed temporarily stored?
 - a. CPU
 - b. Random Access Memory
 - c. Hard Disk
 - d. Read Only Memory
5. Which of the following represent 1st, 2nd, 3rd, and 4th generation programming languages respectively?
 - a. Machine code, assembly code, Visual Basic, C
 - b. Machine language, assembly language, C, SQL
 - c. Machine language, assembly language, SQL, python
 - d. None of the above
6. Choose the incorrect statement(s) from the following.
 - a. Machine Language is a language made up of English like commands.
 - b. Programmers must know a great detail about the computer's design and how it functioned when they are writing programs using machine language.
 - c. Since the language is not machine dependent, assembly languages are easily converted to run on other types of computers.
 - d. Assembler translates the assembly language instruction to programming language instructions.

7. Which option(s) represents the correct sequence of the main phases of the software development life cycle?
- Review specification, informal design, formal design, code, test, maintain the program.
 - Review specification, task list, flow chart, pseudocode, test, maintain the program.
 - Review specification, informal design, algorithm, test, maintain the program.
 - Review specification, informal design, formal design, maintain the program, test.

8. Which of the following statements about desk checking is/are correct?

- It is used to check the logic of the program.
- Step-by-step program design testing.
- Checking the comments in the program.

- a. A only b. A and B c. A and C d. B and C

9. Which is the correct task list for the following problem? "Calculate the average marks of 4 subjects taken separately".

a. set m1, m2, m3 and m4 as mark 1, mark 2, mark 3, and mark 4
set t as total marks
set a as average marks
 $t = m1 + m2 + m3 + m4$
 $a = t/4$
print a

b. set m1, m2, and m3 as mark 1, mark 2, mark 3, and mark 4
set t as total marks
set a as average marks
 $t = m1 + m2 + m3 + m4$
 $t = a/4$
print a

c. set m1, m2, m3 and m4 as mark 1, mark 2, mark 3, and mark 4
set a and t as total and average marks
 $a = m1 + m2 + m3 + m4$
 $a = t/4$
print a

d. set m1, m2, m3 and m4 as mark 1, mark 2, mark 3, and mark 4
set t and a as total marks and average marks
 $t = m1 + m2 + m3 + m4$
 $a = t/4$
print a

10. Which of the following statement(s) correctly defines the information about the loop designed to solve the following problem? Display a table of n and n^2 , for integer values of n ranging from 1 to 10.

- a. Loop control variable: c , initial value of the loop control variable is 1, loop control condition: $c < 10$, repetitive statement: print n and $n \times n$, and $c = c + 1$
- b. Loop control variable: c , initial value of the loop control variable is 1, loop control condition: $c \leq 10$, repetitive statement: print c and $c \times c$, and $c = c + 1$
- c. Loop control variable: c , initial value of the loop control variable is 1, loop control condition: $c < 100$, repetitive statement: print c and $c \times c$, and $c = c + 5$
- d. Loop control variable: c , initial value of the loop control variable is 1, loop control condition: $c < 10$, repetitive statement: print c and $c \times c$, and $c = c + 10$

11. Choose the correct statement(s) after analysing the task list given below.

Until	Teaspoon count for milk powder is 0
Teaspoon	Add a teaspoon of milk
count == 2	Increment the teaspoon count for milk by 1

- a. It performs adding two teaspoons of milk powder into the cup.
- b. It performs adding three teaspoons of milk powder into the cup.
- c. Changing the loop control condition as *teaspoon count* < 3 permits to add three teaspoons of milk powder into the cup.
- d. Changing the initial teaspoon count for milk powder as 2 permit to add two teaspoons of milk powder into the cup.

12. Which of the following is/are false when designing flowcharts?

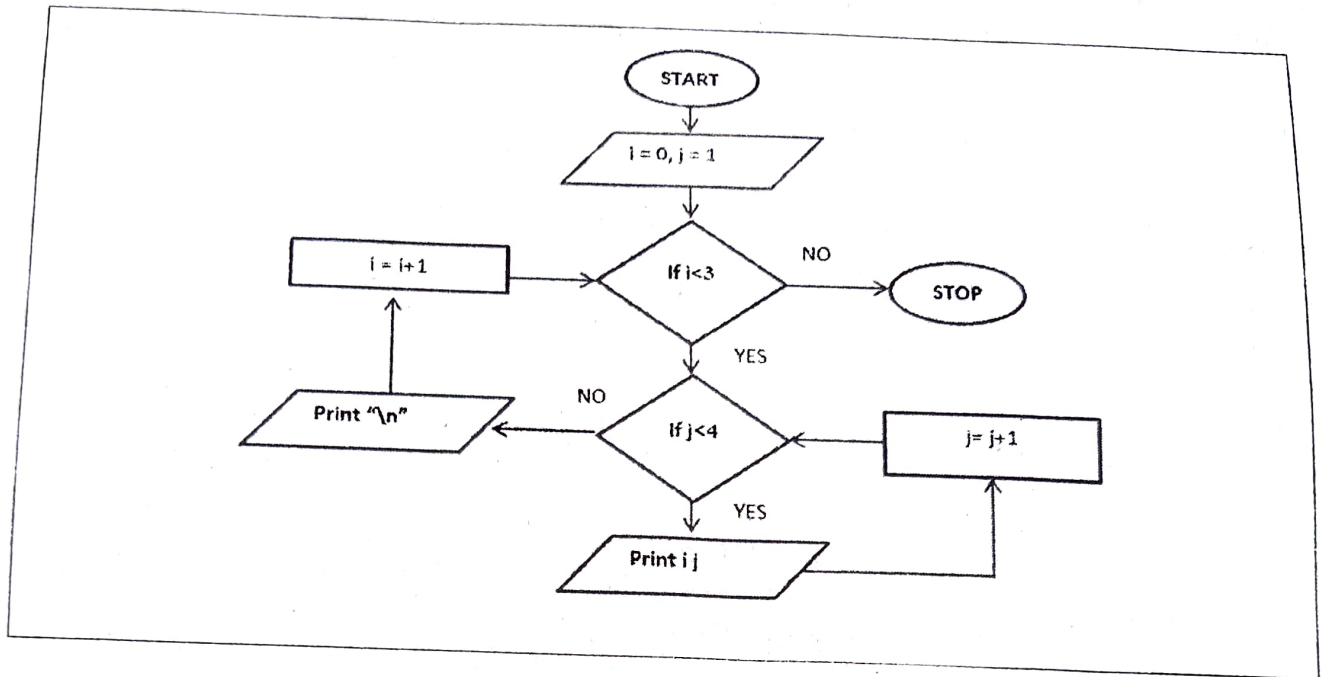
- A. It is designed based on three general design structures namely sequence, selection, and iteration.
- B. Each structure has a single input flow line and multiple exit flow lines.
- C. Flow lines in the flow charts show the flow of control through the program.

- a. A and C
- b. A only
- c. B only
- d. None of the above

13. Which of the following statement(s) correctly explains loops?

- a. Until loop is the leading decision loop, and the while loop is a trailing decision loop.
- b. The body of the until loop will be executed once before control even gets to the loop condition.
- c. In until loop, the loop continues to repeat until the loop condition is true, whereas while loop continues to repeat if the loop condition is true.
- d. In the while loop, if the loop condition is false, control follows the first statement after the loop.

14. Which statement(s) is/are true about the following flowchart?



- a. 010203\n
- b. 010203
- c. 011223
- d. Logic error

15. How to modify the flow chart given in question 14 to obtain the following output?

01020304
11121314
21222324

- a. Change *if j < 4* as *if i <= 4*
- b. Change *if j < 4* as *j <= 4* and add a process block after first decision box to initialize *j = 0*
- c. Remove *print "\n"* output block and change *print ij* output block as *pring ij "\n"*
- d. Add a process block after first decision box to initialize *j = 1*

16. Which statement(s) describes the valid rules followed when creating pseudocode?

- a. Write only one statement per line.
- b. Capitalize initial word.
- c. Indent to show hierarchy.
- d. Keep statements language independent.

17. What will be the output of the following pseudocode, if $n = 5$ and $m = 3$?

```
SET i and j = 1
READ n
READ m

WHILE i < n
    j = 1
    WHILE j < m
        DISPLAY "*"
        j = j + 1
    DISPLAY newline character
    i = i + 1
```

- a. * * * * *
- * * * * *
- * * * * *
- b. * * *
- * * *
- * * *
- * * *
- * * *
- c. * * * *
- * * * *
- d. * *
- * *
- * *
- * *

18. Choose the correct statement(s) from the following.

- a. Sorting the student's marks ascending order is not an example for an algorithm.
- b. A detailed knowledge of hardware is required to write programs using assembly language.
- c. Third generation languages are also named as procedural languages.
- d. Compilers and interpreters are considered as language translators.

19. What this the 5th generation programming language among the following?

- a. Prolog
- b. C
- c. Lisp
- d. Fortran

20. What is meant by a variable in programming?
- a. Variables are used to store information to be referenced and manipulated in a computer program.
 - b. Variable is a keyword in C programming language.
 - c. a. and b.
 - d. none of the above

*** END ***